



Sector:	ICT	Trainer:	Joseph Marie ASHIMIRWE
Trade:	Software Development	School Year:	2025-2026
Qualification Title:	Certificate IV in software Development	Term:	I,II,III
RQF Level:	IV	Module details	
		Module code and title	SWDBD401: DEVELOP A BACKEND APPLICATION USING NODE JS
		Learning hours:	100
		Number of Classes:	1
Date:	12.09.2025	Class Name:	L4 SOD

SCHEME OF WORK

TERM: I

Weeks	Competence code and name: SWDBD401- Develop a Backend Application using Node Js			Learning Activities	Resources (Equipment, tools, and materials)	Evidences of formative assessment	Learning Place	Observation
	Learning Outcome (LO)	Duration	Indicative Content(IC)					
1-7	LO1 : Develop RESTFUL APIs with Node JS	14hrs	IC1.1 : Development environment is properly arranged based on coding architecture methodology	-Group Discussion -Brainstorming -Practical exercise	Internet Laptops Projector Chalks Blackboard Teacher notes VS Code Browser Nodejs	Written Performance	Class room Comp Lab	
			IC1.2 : Server and database connection are properly established according to development environment					
			IC1.3 : RESTFUL APIs are effectively implemented based on backend					

			functionalities					
8-11	LO2 : Secure Backend Application	10hrs	IC2.1 : Data encryption is correctly applied based on system security	-Group discussion -Blainstorming -Group exercises	Internet Projector Power point Hand out notes Chalks Blackboard Laptops VS Code Browser Nodejs	Written Performance	Computer lab Classroom	
			IC2.2 : Third-party libraries are carefully checked based on system security					

TERM: II

Weeks	Competence code and name: SWDBD401- Develop a Backend Application using Node Js			Learning Activities	Resources (Equipment, tools, and materials)	Evidences of formative assessment	Learning Place	Observation
	Learning Outcome (LO)	Duration	Indicative Content(IC)					
1-4	LO2 : Secure Backend Application	10hrs	IC2.3 : User Authentication, Authorization and	-Group discussion -Blainstorming	Internet Projector Power point	Written Performance	Computer lab	

			Accountability (AAA) are carefully applied based on NPM Universal Access Control (UAC)	-Group exercises	Hand out notes Chalks Blackboard VS Code Browser Nodejs			
			IC2.4: Environment variables are carefully Secured according to system security					
5-7	LO3 : Test Backend Application	20hrs	IC3.1: Unit tests are appropriately conducted based on software testing techniques	-Group discussion -Blainstorming -Group exercises -Lab Practices	Internet Projector Power point Hand out notes Chalks Blackboard VS Code Browser Nodejs	Written Perfomance	Computer lab	
			IC3.2: Usability is correctly tested according to expected results					
			IC3.3: Security is properly tested based on system threats					

TERM: III

Weeks	Competence code and name: SWDBD401- Develop a Backend Application using Node Js			Learning Activities	Resources (Equipment, tools, and materials)	Evidences of formative assessment	Learning Place	Observation
	Learning Outcome (LO)	Duration	Indicative Content(IC)					
5-8	LO4 : Manage Backend Application	15hrs	IC4.1: Application is appropriately deployed based on FURPS requirements	-Group discussion -Blainstorming -Group exercises -Lab practice	Internet Projector Power point Hand out notes Chalks Blackboard VS Code Browser Nodejs			
			IC4.2: Backend is effectively maintained according to the system Functionalities					
			IC4.3: Application documentation is properly generated according to the system backend					

9	Integrated Assessment (for specific module)	To develop a web application using Node.js and MySQL	Consumables Internet Equipments Computers Tools Installed Software editor Flash disk	Practical assesement	Computer Lab	
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Prepared by: Trainer

Joseph Marie ASHIMIRWE

Verified by: DOS

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Approved by: SM

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